RESEARCH CALIBRATION & METHODOLOGY CONSOLIDATION

Passenger Price Sensitivity & NYC Fare Calibration

Consolidated Explanation of Meeting Recordings (5 & 5(2)) Integrated with Empirical Dataset Array Calculations

Sources: AudioCutter Standard Recording 5(2) (5-page transcript), AudioCutter Standard Recording 5 (6-page transcript), and Empirical Dataset Array a (148 distance bins, L = 1.0 to 74.5).

Document Purpose & How to Read It

This document consolidates the theoretical and methodological instructions from both meeting transcripts alongside the concrete empirical calculations derived from the 148-row NYC trip dataset array a. Recording 5(2) addresses the demand/distance distribution, mean fare, passenger price sensitivity (ε / e), fitted sensitivity distribution, price bounds, P_{max}, pricing actions, and simulation execution. Recording 5 addresses the quality of ε estimates, NYC fare-distance data validation, taxi-meter base fare and base distance rules, distance-bin filtering, minimum observation counts, and the proposed distance range.

Throughout this document, **actual empirical results computed from your dataset array a** (such as the count-weighted mean price of 22.3542, nearest base distance of 4.5, and the complete 148-element sensitivity array e) are presented explicitly as empirical dataset outputs, separated from illustrative examples used in the transcripts.

> **Distinction Between Empirical Calculations, Source Directives, and Illustrative Examples:**
> 1. **Empirical Dataset Outputs (Section 1):** Exact values calculated directly from your 148-row matrix a (Columns: Distance L, Trip Count N, Price P).
> 2. **Source-Derived Directives:** The professor's instructions and observations from Recordings 5 and 5(2).
> 3. **Illustrative Examples:** Placeholder tables and numbers from the transcript explanation (explicitly marked as *example*) that must not replace actual dataset outputs.

Executive Summary

The central objective is to build a statistically calibrated passenger population for the dynamic-pricing simulation. Rather than relying on a single average elasticity or uniform trip assumptions, the simulation must generate trips using the observed NYC distance/demand distribution, apply a defensible distance–fare relationship and base-fare floor, assign heterogeneous passenger price sensitivities sampled from a fitted distribution, and constrain generated prices within empirically supported NYC lower and upper bounds.

- **Empirical Dataset Baseline:** Across the 148 distance bins (L = 1.0 to 74.5, total trip count = 7,946,921), the count-weighted mean price is **22.3542**, and the nearest observed distance bin to that mean price is $\boldsymbol{L}$ **= 4.5** (observed price = 21.9832).
- **Empirical Sensitivity Array (e):** Using P_{base} = 22.3542 and L_{base} = 4.5, the computed sensitivity array e across all 148 bins yields a sample mean of **2.313938**, a sample standard deviation of **0.408834** (population SD: 0.407451), a minimum of **0.000000** (at L = 4.5), and a maximum of **2.938795** (at L = 45.5).
- **Population Sampling Over Deterministic Lookup:** Treat ε as a population probability distribution rather than assigning a bin's deterministic e value to every simulated trip in that distance bin.
- **Data Hygiene & Bounds:** Investigate extreme ε outliers (such as values near 500 or 1,000 in raw trip records), distinguish taxi-meter base distance/fare from raw minimum records ($0.01), retain bins up to ~75 km with adequate counts (50–100+ trips), and clip simulated prices to $[P_{NYC,lower}, P_{NYC,upper}]$.

PART I — EMPIRICAL DATASET CALIBRATION & SENSITIVITY ARRAY (ARRAY A)

1. Dataset Structure & Weighted Mean Price Calibration

The empirical dataset array a consists of 148 uniformly spaced distance bins (L from 1.0 to 74.5 in increments of 0.5), where Column 1 represents trip distance (L), Column 2 represents trip count (N), and Column 3 represents observed price (P). Total observed trips across all 148 bins equal **7,946,921** (matching the ~8 million sample size discussed by the professor for the ~75 km cutoff).

> **Count-Weighted Mean Price:**
> $P_{mean} = (\sum_{i=1}^{148} N_i \times P_i) / (\sum_{i=1}^{148} N_i)$ = 177,647,049.04 / 7,946,921 = **22.35419844 ≈ 22.3542**

To establish the reference base distance (L_{base}) corresponding to P_{base} = 22.3542, the surrounding distance bins in array a are compared below. Bin $\boldsymbol{L}$ **= 4.5** is the closest discrete distance to the count-weighted mean price (linear interpolation between 4.5 and 5.0 yields 4.6170).

Distance (L)	Trip Count (N)	Observed Price (P)	Abs. Difference from P_{base} (22.3542)	Selection Status
4.0	364,612	20.5232	1.8310	Adjacent lower bin
4.5	**310,535**	**21.9832**	**0.3710**	**Selected nearest L_{base} = 4.5**
5.0	240,561	23.5689	1.2147	Adjacent upper bin

2. Empirical Sensitivity Array (e) Formula & Summary Statistics

Setting P_{base} = 22.3542 and L_{base} = 4.5, the sensitivity array e is computed for every bin in array a using the relative elasticity formulation:

> $e_i = [\ (L_i - L_{base}) / (P_i - P_{base})\ ] \times (P_{base} / L_{base}) = [\ (L_i - 4.5) / (P_i - 22.3542)\ ] \times (22.3542 / 4.5)$

Across all 148 rows of array a, the resulting sensitivity vector e exhibits the following summary statistics (aligning closely with the ~2.31 mean and zero-value bin noted in Recording 5(2)):

Statistic	Computed Value	Notes & Observations
Mean (μ_e)	**2.313938**	Unweighted arithmetic mean across all 148 distance bins
Sample Standard Deviation (s_e, $N-1$)	**0.408834**	Standard Bessel-corrected dispersion across the 148 bins
Population Standard Deviation (σ_e, N)	**0.407451**	Population dispersion (N = 148)
Minimum (e_{min})	**0.000000**	Occurs at L = 4.5 where $L - L_{base} = 0$ ($P = P_{base}$ rule)
Maximum (e_{max})	**2.938795**	Occurs at L = 45.5 (N = 786, P = 91.6587)
Short-Distance Regime (L = 1.0–4.0)	1.3220 – 1.3858	Lower sensitivity values below L_{base} = 4.5
Mid-to-Long Regime (L = 5.0–74.5)	1.7075 – 2.9388	Higher variance at L > 60 where bin counts drop below 100

3. Complete Empirical Dataset & Calculated Sensitivity Array (e)

The table below records all 148 bins of dataset array a (L, N, P) paired with the calculated sensitivity value e for each bin.

L	Count (N)	Price (P)	e	L	Count (N)	Price (P)	e
1.0	979,766	9.202250	1.321979	38.0	1,242	86.032979	2.613345
1.5	993,072	11.227201	1.339337	38.5	1,167	86.818218	2.620041
2.0	872,323	13.223929	1.360201	39.0	1,118	87.281655	2.639595
2.5	723,437	15.121041	1.373563	39.5	1,052	89.830076	2.576713
3.0	580,862	16.977309	1.385820	40.0	1,074	91.672765	2.544049
3.5	461,472	18.758150	1.381404	40.5	1,163	90.206131	2.635645
4.0	364,612	20.523164	1.356500	41.0	1,215	92.997078	2.566676
4.5	310,535	21.983204	0.000000	41.5	1,370	93.547752	2.581711
5.0	240,561	23.568871	2.044833	42.0	1,577	95.545048	2.545195
5.5	200,866	24.828967	2.007300	42.5	1,387	97.243526	2.520637
6.0	170,408	26.021931	2.031610	43.0	1,312	99.697934	2.472761
6.5	145,746	27.168103	2.063855	43.5	1,197	97.556048	2.576219

L	Count (*N*)	Price (*P*)	*e*	*L*	Count (*N*)	Price (*P*)	*e*
7.0	130,974	28.284578	2.094133	44.0	1,045	94.794766	2.708706
7.5	113,910	29.272057	2.154251	44.5	969	94.795026	2.742984
8.0	101,390	30.316726	2.183553	45.0	954	92.618229	2.863311
8.5	90,537	31.280076	2.226157	45.5	786	91.658651	2.938795
9.0	86,403	32.008001	2.315585	46.0	688	97.449869	2.745237
9.5	74,345	32.933630	2.347764	46.5	574	94.424913	2.894923
10.0	68,529	33.793947	2.388322	47.0	462	100.716515	2.694190
10.5	64,270	34.669203	2.420267	47.5	456	105.466579	2.570096
11.0	59,685	35.513799	2.453677	48.0	399	108.015739	2.522609
11.5	55,136	36.344433	2.485534	48.5	316	112.238228	2.431738
12.0	51,518	37.250730	2.501052	49.0	335	117.068388	2.333950
12.5	50,427	37.824466	2.568850	49.5	287	117.991045	2.337405
13.0	50,343	38.534619	2.609611	50.0	235	121.573702	2.278038
13.5	51,439	39.409967	2.621307	50.5	241	120.688963	2.323793
14.0	54,236	40.272803	2.633699	51.0	249	128.214538	2.182058
14.5	50,393	41.468863	2.598843	51.5	209	130.912967	2.150699
15.0	46,444	42.252807	2.621279	52.0	196	130.560663	2.180655
15.5	42,787	43.181799	2.623615	52.5	184	125.487011	2.312017
16.0	39,506	44.134473	2.622896	53.0	190	133.308526	2.171421
16.5	35,525	45.022267	2.629743	53.5	159	129.288994	2.276269
17.0	32,900	45.844759	2.643402	54.0	149	135.061745	2.181719
17.5	28,982	46.773474	2.644583	54.5	170	127.889235	2.353531
18.0	26,046	47.822773	2.633151	55.0	143	142.844056	2.082033
18.5	26,558	48.244739	2.686170	55.5	159	145.555975	2.056363
19.0	20,642	49.564612	2.647156	56.0	155	141.898516	2.140055
19.5	16,913	50.817403	2.617906	56.5	127	146.041102	2.088460
20.0	14,721	51.762633	2.618222	57.0	147	149.808095	2.046222
20.5	13,480	52.923680	2.600031	57.5	160	149.502688	2.070672
21.0	12,434	53.458420	2.635186	58.0	135	162.409556	1.897583
21.5	11,859	54.850360	2.598744	58.5	164	160.300793	1.944596
22.0	10,962	55.979820	2.585320	59.0	140	157.744214	1.999662
22.5	10,770	56.606140	2.610562	59.5	126	158.686349	2.004061
23.0	12,167	57.126409	2.642932	60.0	108	158.509444	2.024908

L	Count (*N*)	Price (*P*)	*e*	*L*	Count (*N*)	Price (*P*)	*e*
23.5	10,617	59.090286	2.569256	60.5	123	157.136098	2.063969
24.0	10,655	60.425859	2.544365	61.0	83	161.870723	2.011729
24.5	10,558	61.965298	2.508186	61.5	110	173.841636	1.869153
25.0	11,405	63.571446	2.470709	62.0	99	150.753030	2.224607
25.5	12,957	65.575220	2.413631	62.5	97	167.767113	1.981398
26.0	15,414	67.288400	2.376884	63.0	79	167.396835	2.003581
26.5	18,775	68.499687	2.368318	63.5	91	182.047692	1.835318
27.0	22,358	69.321955	2.379739	64.0	67	178.512090	1.892778
27.5	24,662	69.895721	2.403263	64.5	79	168.149494	2.044346
28.0	26,239	69.885766	2.456023	65.0	73	178.910274	1.919694
28.5	23,205	70.325524	2.485285	65.5	74	163.518108	2.146608
29.0	20,688	70.682516	2.518321	66.0	64	169.644844	2.074181
29.5	17,696	70.687918	2.569428	66.5	57	184.491754	1.899567
30.0	15,074	71.082894	2.599573	67.0	57	187.177368	1.883685
30.5	12,757	71.333602	2.636978	67.5	62	190.566129	1.860503
31.0	10,664	72.102078	2.646171	68.0	48	207.088542	1.707547
31.5	9,147	72.203525	2.690612	68.5	64	176.468438	2.062927
32.0	8,559	73.112149	2.691381	69.0	50	199.065800	1.813181
32.5	7,926	72.489871	2.774328	69.5	53	176.883208	2.089536
33.0	6,266	73.847434	2.749421	70.0	56	187.590179	1.969170
33.5	5,135	73.556378	2.813560	70.5	46	166.760870	2.270405
34.0	4,081	75.228493	2.771559	71.0	55	193.079818	1.934949
34.5	3,316	75.599897	2.798874	71.5	37	187.885946	2.010667
35.0	2,676	76.558632	2.795192	72.0	36	183.863611	2.076121
35.5	2,197	77.364210	2.799410	72.5	52	185.560769	2.069750
36.0	1,948	79.644348	2.731349	73.0	56	190.033750	2.029351
36.5	1,583	79.938124	2.760548	73.5	50	200.387400	1.925284
37.0	1,583	79.928339	2.804158	74.0	45	166.318222	2.398156
37.5	1,300	84.158192	2.652431	74.5	35	177.286286	2.244416

PART II — RECORDING 5(2): DEMAND, SENSITIVITY MODELING & SIMULATION

4. Demand & Distance Distribution

The professor examines how trip demand varies with distance. While noting that the empirical distribution resembles a Zipf-type or exponential-type decay (with secondary peaks visible in array a around L = 14.0, 28.0, and 42.0), the practical directive is to restrict trip generation to the empirical distribution observed in the dataset rather than generating distances uniformly.

> **Normalized Distance-Bin Probability:** $\Pr(\text{distance bin } i) = N_i / \sum_j N_j$ (where $\sum_i p_i = 1$)

Pickup and drop-off grid locations should similarly be sampled according to the spatial grid frequencies observed in the NYC dataset.

5. Meaning of Passenger Price Sensitivity (ε) & Zero-Sensitivity Rule

Passenger price sensitivity (ε) quantifies how strongly a passenger responds to a price change: lower ε indicates lower price sensitivity, whereas higher ε indicates stronger sensitivity. Consistent with the transcript discussion and the empirical calculation at L = 4.5 (e = 0.000000), when sensitivity is zero, no sensitivity-driven price adjustment occurs:

> **Zero-Sensitivity Condition:** If $\varepsilon = 0$, then $P = P_{base}$

Illustrative Passenger (Example)	Illustrative ε	Plain-Language Interpretation
Passenger A (Example)	0.5	Relatively less price-sensitive
Passenger B (Example)	1.0	Moderate sensitivity
Passenger C (Example)	2.0	Higher sensitivity (near dataset mean $\mu_e \approx 2.31$)
Passenger D (Example)	4.0	Very high sensitivity

6. Treating ε as a Population Distribution & Distribution Fitting

A core instruction in Recording 5(2) is **not** to force a simulated passenger to use the deterministic e recorded for that exact distance bin. Instead, treat ε as a population probability distribution so that simulated passengers across different distances draw heterogeneous sensitivity values from a common fitted distribution.

Because the earlier assumption of a normal distribution does not match the skewed empirical observations, the professor directs the team to fit a suitable probability distribution using the empirical mean (μ_e = 2.313938) and standard deviation (s_e = 0.408834), testing exponential-type and related candidates rather than assuming exponential without validation:

- Inspect raw ε observations (including zero, negative, near-zero, and extreme values) and plot histograms and empirical CDFs.
- Compute summary statistics (mean, standard deviation, min, max, median, quantiles, and skewness).
- Fit candidate distributions supported by the data (e.g., exponential, gamma, log-normal, Weibull, alongside normal as a baseline comparison).
- Compare fits using histogram overlays, Q–Q plots, AIC/BIC criteria, and Kolmogorov–Smirnov goodness-of-fit tests, documenting any nonnegativity support ($\varepsilon \geq 0$) or truncation rules.

7. P_{base}, NYC Price Bounds, P_{max}, Clipping & Action Space

The simulation distinguishes the reference base price (P_{base}) from empirical NYC lower and upper fare limits ($P_{NYC,lower}$, $P_{NYC,upper}$) and the model's theoretical maximum generated price (P_{max}). Any generated price falling outside the NYC bounds must be clipped to the boundary:

> **Price Truncation / Clipping Rule:** $P_{final} = \min(P_{NYC,upper}, \max(P_{NYC,lower}, P_{generated}))$

Parameter / Concept	Definition & Calibration Rule
P_{base}	Reference/base fare from finalized fare calibration (count-weighted mean price = **22.3542** in array *a*).
$P_{NYC,lower}$ & $P_{NYC,upper}$	Empirical NYC lower and upper fare limits (e.g., illustrative $7.52 and $207 discussed in transcript; verify against cleaned dataset—note max bin price in array *a* is 207.09 at *L* = 68.0).
P_{max}	Maximum price generated by the model, calculated via the approved pricing formulation and calibrated inputs.
Pricing Action Space (a)	Multiplicative pricing factors $P_{action} = P_{base} \times a$ (e.g., illustrative set {0.85, 0.90, 0.95, 1.00, 1.05, 1.10, 1.15}). Inspect action selection frequencies to verify that long-distance rarity does not starve upper actions.
Weather & Time Conditioning	Initially use one pooled ε distribution and one common P_{base} across clear/rain weather and time slots; evaluate weather-wise and time-wise splits as secondary experiments.

PART III — RECORDING 5: DATA VALIDATION & BASE FARE CALIBRATION

8. Investigation of Extreme ε Values (500 – 1,000)

In Recording 5, the professor questions extreme ε values around 500 and 1,000 observed in earlier plots (when earlier summaries suggested means around 0.6–0.7 and dispersion around 6–7). Because elasticity-style ratios blow up when the denominator approaches zero (for instance, a 10% numerator change divided by a 0.01% price difference equals 1,000), extreme values must not be blindly deleted without inspection:

- Retrieve source records exceeding an investigation threshold and inspect trip IDs, raw fares, distances, and numerator/denominator differences.
- Identify near-zero denominators, duplicate records, mismatched pairings, or sign-convention discrepancies (signed vs. absolute vs. positive magnitude).
- Document an evidence-based decision (retain, correct, or exclude) for every flagged record.

9. Taxi-Meter Base Fare vs. Minimum Observed Fare

The professor rejects using raw minimum dataset records ($0.01 fare or 0.016 km distance) as the minimum fare floor. Under taxi-meter pricing, passengers pay a fixed base fare (P_{base}) for any trip up to the base distance (L_{base}), after which fare scales with distance. Crucially, the raw observed distance ($L_{observed}$) must be preserved for demand and routing analytics while a separate effective distance ($L_{effective}$) is used for fare calculation:

> **Effective Distance & Base-Fare Rule:** $L_{effective} = \max(L_{observed}, L_{base})$ | If $L \le L_{base}$, $P = P_{base}$

10. Distance Range Cutoffs, Minimum Bin Counts & Required Table

To eliminate sparse tail noise and anomalous short-trip records, the professor discusses starting the usable fare-distance range around 1 km (also considering 1.5–2 km), capping the upper distance around **75 km** (which retains ~8 million observations, consistent with the 7,946,921 trips in array *a* up to 74.5), and enforcing a minimum observation threshold of **50 or 100 trips per bin** (note that in array *a*, bins from L = 61.0 to 74.5 drop below 100 trips, reaching 35 trips at L = 74.5, which explains the increased price noise in that tail).

Selection Parameter	Professor's Discussion	Empirical Array *a* & Implementation Status
Lower Distance Cutoff	Around 1 km (1.5 or 2 km also discussed)	Array *a* starts at L = 1.0 (N = 979,766); compare candidate cutoffs.
Upper Distance Cutoff	~75 km selected for practical dataset (~100 km discussed)	Array *a* spans up to L = 74.5, retaining **7,946,921** total trips.
Minimum Bin Count	50 or 100 trips per distance interval	All bins ≤ 60.5 exceed 100 trips; bins 61.0–74.5 range from 35 to 110 trips.
Required Bin Fields	Distance bin, trip count, and mean fare	Supported by median fare, fare SD, and percentiles (P5, P25, P75, P95).

PART IV — END-TO-END SIMULATION WORKFLOW, CHECKLIST & CAUTIONS

11. Request-by-Request Simulation Execution

1. **Sample Distance Bin & Continuous Distance:** Draw a distance bin from the normalized empirical distribution $p_i = N_i / \sum N_j$, then sample an exact trip distance within that bin.
2. **Assign Spatial Grid Coordinates:** Sample pickup and drop-off grid cells from the empirical NYC spatial distribution subject to feasible OD routing rules.
3. **Sample Passenger Sensitivity (ε):** Draw an independent ε value from the validated population distribution fitted to the cleaned empirical sensitivity observations.
4. **Compute Reference Fare & Pricing Action:** Apply the base-distance floor ($L_{effective} = \max(L_{observed}, L_{base})$), compute the reference fare, and apply the pricing model's selected action ($P_{action} = P_{base} \times \alpha$).
5. **Clip Price & Evaluate Response:** Clip $P_{generated}$ to $[P_{NYC,lower}, P_{NYC,upper}]$, evaluate passenger acceptance using the approved utility/response model, and log all outputs for validation against empirical distributions.

12. Prioritized Implementation Checklist & Deliverables

☐ Audit extreme ε records (values ~500–1,000) and verify sign convention, denominator stability, and units.

☐ Finalize distance-bin cutoffs (~1 km to ~75 km) and test minimum bin-count thresholds of 50 vs. 100 trips (evaluating tail bins $L \geq 61.0$).

☐ Verify fare column definitions, resolve repeated \$0.01 entries, and finalize L_{base}, P_{base} (22.3542 in array a), and NYC price limits $[P_{NYC,lower}, P_{NYC,upper}]$.

☐ Fit and compare candidate probability distributions (exponential, gamma, log-normal, Weibull) to the validated ε dataset using histograms, empirical CDFs, Q–Q plots, and AIC/BIC.

☐ Compute P_{max} from the approved pricing formulation, verify action-space coverage, implement price clipping, and validate simulated outputs against empirical NYC distributions.

13. Required Diagnostic Plots & Unresolved Items

Category	Specific Items & Cautions
Required Visualizations	(1) Trip distance histogram/probability plot; (2) Trip count by distance bin; (3) Distance vs. mean fare curve & boxplots; (4) Sensitivity (ε) histogram, empirical vs. fitted CDF, and Q–Q plots; (5) Pre- vs. post-clipping price distributions and action selection frequencies.
Unresolved Parameters to Confirm	Final L_{base} (e.g., 1 km meter threshold vs. 4.5 mean-price bin), exact $P_{NYC,lower}$ and $P_{NYC,upper}$ bounds, complete P_{max} formula, final parametric distribution for ε, and secondary weather/time-of-day stratification.
Timeline Note	In Recording 5(2), the professor targets completing sensitivity calibration within November (worst-case end of December) so the remaining simulation pipeline can execute promptly.